LECTURE 1 — STRESS, PATHOLOGICAL & OPEN FRACTURES

Stress (fatigue) fractures

- Fracture from **repetitive submaximal stress** on normal bone (not single trauma).
- **At risk**: athletes, military recruits, dancers, steroid users, **females with irregular menses** (female athlete triad), osteoporosis.
- **Imaging**: plain X-ray (often normal early), CT, **MRI**, **radioisotope (bone) scan** (sensitive early). [Common sites: metatarsals (march fracture), tibia, femoral neck — standard.]
- **Treatment**: rest + activity modification, **muscle strengthening** (calves/shin to dissipate forces), bracing/air-cast boot, crutches; severe → surgery (pinning), rehab up to 6 months.

Pathological fractures

- Fracture through abnormal/weakened bone. **Biopsy = keystone of diagnosis** (closed: trucut/wide-pore needle; open: incisional/excisional — marginal/wide-local/radical).

Through benign tumours

- **More common than through malignant**; mostly asymptomatic before fracture; children; humerus + femur. Causes: **unicameral bone cyst (UBC), non-ossifying fibroma (NOF), fibrous dysplasia, eosinophilic granuloma**.
- **Unicameral (simple) bone cyst**: males > females; first 2 decades; humerus/femur; **"fallen fragment" sign**. Treat: observe / aspiration + methylprednisolone injection / curette + graft; **ORIF for femoral neck**.
- **Fibroxanthoma (NOF)**: most common benign; femur/distal tibia/humerus; multiple in 8% (neurofibromatosis); ↑fracture risk if **>50% bone diameter / >22mm**. Treat: observe / curette + graft if impending.
- **Fibrous dysplasia**: solitary (commonest) or multifocal; femur/humerus; **café-au-lait + endocrinopathy = Albright's syndrome**. Treat: observe / curette + graft, **bisphosphonates**.

Through primary malignant tumours

- Rare, often unsuspected. **Osteosarcoma, Ewing's, MFH, fibrosarcoma**. Suspect in **younger patients + aggressive lesion** (poorly defined margins, wide transition zone, matrix production, periosteal reaction); **antecedent night pain**.
- Fracture doesn't preclude limb salvage (↑local recurrence but survival not compromised). **Always biopsy solitary destructive lesions** even with carcinoma history. Treat: immobilize, stage, biopsy, chemo, resection/amputation.

Through metabolic/non-neoplastic bone disease

- **Osteoporosis** (insufficiency fractures), **Paget's** (most fractures in late stage; **excessive bleeding**, delayed healing, malignant transformation), **hyperparathyroidism** (fractures through **brown tumours**).
- **Paget's**: thickened cortices, mixed sclerosis/lysis, bowing → osteotomy/arthroplasty.
- **Hyperparathyroidism**: adenoma; mental change, abdominal pain, nephrolithiasis; **brown tumours**; treat parathyroidectomy + ORIF + correct calcium.

Through metastatic disease & myeloma

- After osteoporosis, **commonest cause** of pathological fracture; 5th decade+; **femur + humerus**.
- **Not all mets equal**: breast/prostate/thyroid = radio + chemosensitive; **renal = minimally radiosensitive** (embolize pre-op).

- **61% of pathological fractures in femur (80% peritrochanteric)**.
- **Harrington criteria for prophylactic fixation**: lesion **>50% bone diameter**, **>2.5 cm**, **pain after radiation**, **lesser trochanter avulsion fracture**.
- **Adjuvant**: radiation (85% response, pain relief, recalcification 2–3mo), bisphosphonates (pamidronate/zoledronic acid), chemo/hormones.
- **Treatment**: biopsy solitary lesions; nail vs plate vs arthroplasty (**plates+cement best for torsion; interlocked nail stabilizes whole bone**); **arthroplasty for periarticular/renal/failed fixation/post-radiation**; immediate weight-bearing goal.

Open fractures

- **All open fractures = contaminated. 4 essentials**: immediate wound cover, antibiotic prophylaxis, early debridement, fracture stabilization.
- **4 questions**: nature of wound? state of surrounding skin? circulation? nerves intact?
- **Gustilo classification**:
- **I**: <1cm clean puncture, minimal soft tissue damage, not comminuted.
- **II**: >1cm, no flap, moderate crush/comminution.
- **III**: extensive damage + contamination. **IIIA** = bone coverable; **IIIB** = periosteal stripping + severe comminution (needs flap); **IIIC** = **arterial injury needing repair**. (High-velocity = IIIB/C.)
- Infection risk rises with grade (<2% type I → >10% type II+).
- **Early management**: keep covered, **antibiotics ASAP** (benzylpenicillin + flucloxacillin 6-hourly ×48h; add gentamicin/metronidazole if contaminated; **tetanus prophylaxis**). **Debridement** (no tourniquet — see devitalized tissue), copious saline irrigation.
- **Wound closure**: small clean type I → primary suture/graft; **all others left open** → **inspect at 5 days** → **delayed primary closure**.
- **Stabilization**: type I/small II + stable → plaster/traction; severe + gunshot → fix. **External fixation = safest**; IM nailing (femur/tibia); plates for metaphyseal/articular.

Lecture 1 — high-yield

- **Stress fracture = repetitive stress, normal bone; bone scan/MRI early; female athlete triad.**
- **UBC = fallen fragment sign; fibrous dysplasia = Albright's (café-au-lait + endocrinopathy).**
- **Always biopsy solitary destructive lesions. Renal mets = embolize (poor radiosensitivity).**
- **Harrington criteria: >50% diameter, >2.5cm, pain after radiation, lesser trochanter fracture.**
- **Open fracture = contaminated; Gustilo I–IIIC (IIIC = arterial injury); antibiotics + debridement + ext fix; closure delayed.**

LECTURE 2 — BONE TUMOURS (Radiology)

Imaging approach

- **Plain film = best for diagnosis.** MRI/CT = full extent, soft tissue, neurovascular relationship (**MRI > CT** for local extent). **Isotope scan = metastatic disease.**
- **Metastases = by far the commonest bone neoplasm.**
- **Assess by: (1) morphology** (well-defined lytic / ill-defined lytic / sclerotic) **and (2) age**.
- **Ill-defined zone of transition** = aggressive: **malignant** (Ewing's, osteosarcoma, leukaemia, mets, myeloma) or aggressive benign (infection, eosinophilic granuloma, GCT).
- **Well-defined** = usually benign (<30 yrs); but **>40 yrs include mets + myeloma**.
- **Benign signs**: narrow transition + sclerotic rim, expansion but rarely breaches cortex, no soft tissue mass, no periosteal reaction (unless fracture), little isotope uptake.
- **Malignant signs**: poorly defined margin, wide transition zone, **cortical destruction, periosteal reaction, soft tissue mass**, ↑isotope activity.

Primary malignant bone tumours

Tumour	Age	Site	X-ray
Osteosarcoma	5–20 (+ elderly Paget's)	Metaphysis, **around knee**	Bone destruction + new bone, **"sunray/sunburst" spiculated periosteal reaction, Codman's triangle**
Chondrosarcoma	30–50	Pelvis, scapula, humerus, femur	Lytic expansile + **flecks of calcium** (hard to distinguish from enchondroma)
Ewing's sarcoma	5–15	Long bone **diaphysis**	Ill-defined destruction + **"onion-skin" lamellated periosteal reaction**, permeative
Giant cell tumour	20–40	**Around knee/wrist, subarticular (epiphysis)**	Expanding lytic, well-defined margin, thin/destroyed cortex; locally invasive, rarely metastasizes

Benign tumours & tumour-like conditions

- **Enchondroma**: lytic expanding, **hand**, chondroid matrix (calcium flecks), often pathological fracture.
- **Fibrous cortical defect / NOF**: incidental in children; well-defined lucent cortical, scalloped sclerotic margin.
- **Fibrous dysplasia**: one/several bones, long bones + ribs; lucent, well-defined, may expand, **"ground glass"**.
- **Simple bone cyst**: children, humerus/femur, lucency across shaft, **fallen fragment sign**, presents as fracture.
- **Aneurysmal bone cyst**: not a true neoplasm; children; spine/long bone/pelvis; **purely lytic, massive "aneurysmal" expansion**; CT/MRI fluid-fluid levels; DDx = GCT.
- **Osteoid osteoma**: **painful (night, relieved by NSAIDs)**, femur/tibia, young adults; **lucent nidus <2cm + dense sclerotic rim**; bone scan very hot.

Multiple focal lesions

- **Metastases = commonest malignant bone tumour**; red marrow sites (spine, skull, ribs, pelvis, humeri, femora).
- **Lytic**: adults — breast, bronchus, thyroid, renal, colon; children — neuroblastoma, leukaemia.
- **Sclerotic**: **men = prostate; women = breast**. Mixed = breast.

- Expansion uncommon (except thyroid/kidney); **isotope > plain film; MRI > isotope; CT < MRI; FDG-PET/CT for staging**.
- **Multiple myeloma**: commonest primary malignant bone tumour in adults; active haemopoiesis bones; lytic (better defined than mets, may expand) or osteoporosis-like; **MRI very effective**.
- **Multiple periosteal reactions**: non-accidental injury, congenital syphilis, venous stasis, **hypertrophic pulmonary osteoarthropathy**, scurvy.

Decreased bone density (osteopenia)

- Causes: **osteoporosis, osteomalacia, hyperparathyroidism, multiple myeloma**.
- **Osteoporosis**: ↓protein matrix (osteoid), normal mineralization; → vertebral + hip fractures. Causes: idiopathic (juvenile/senile/**postmenopausal — 50% women >60**), Cushing's/steroids, disuse. X-ray: ↓density, thin cortex, ↓trabeculae; spine — **"empty box"/picture frame, cod-fish vertebrae**, compression fracture. **DEXA: osteoporosis = T-score** ≤ **–2.5**.
- **Rickets (before epiphyseal closure) / osteomalacia (adult)** = poor osteoid mineralization. Rickets: **widened physis, cupped/frayed metaphysis, rachitic rosary**, greenstick fractures (best at knees/wrists/ankles). Osteomalacia: **Looser's zones** (lucent bands — scapula, femur, pubic rami, ribs), biconcave vertebrae.
- **Hyperparathyroidism**: ↓density, **subperiosteal bone resorption** (radial phalanges), soft tissue/chondrocalcification, **brown tumours**. Primary (90% adenoma) vs secondary (renal).

Increased bone density

- **Sclerotic metastases (commonest)**, **osteopetrosis** (marble bone, congenital, multiple fractures), myelosclerosis.

Trabecular pattern / shape changes

- **Paget's disease**: elderly chance finding; pelvis/spine/skull/long bones; **thickened trabeculae + cortex, loss of corticomedullary differentiation, bone enlargement, bowing**; picture-frame vertebra. Complication: fracture, malignant transformation.
- **Haemolytic anaemia (thalassaemia/sickle)**: marrow hyperplasia → thin cortex, bone expansion, **"hair-on-end" skull**, rectangular phalanges; sickle → infarction + infection.
- **Diaphyseal aclasia (multiple exostoses)**: congenital; multiple osteochondromas near metaphyses, **pointing away from joint**, cartilaginous cap.

Lecture 2 — high-yield

- **Plain film best for dx; mets = commonest bone neoplasm; assess morphology + age.**
- **Osteosarcoma = knee, sunburst + Codman's; Ewing's = diaphysis, onion-skin; chondrosarcoma = calcium flecks; GCT = subarticular epiphysis around knee.**
- **Osteoid osteoma = painful nidus <2cm, relieved by NSAIDs. UBC/SBC = fallen fragment. ABC = aneurysmal expansion (DDx GCT).**
- **Sclerotic mets: men prostate, women breast. Myeloma = MRI; lytic well-defined.**
- **Osteoporosis = DEXA T ≤ –2.5, normal mineralization; osteomalacia = Looser's zones; rickets = cupped/frayed metaphysis.**
- **HyperPTH = subperiosteal resorption + brown tumours. Paget's = thick trabeculae + bone enlargement.**